Mounir Ammam

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EDUCATION

M.Sc. in Machine Learning — Montreal, QC
McGill University & Mila — Starting Sept 2026 (Expected: Dec 2027)

Bachelor of Science in Computer Science (*Honours*) — Montreal, QC
Université de Montréal — Graduation (Expected): April 2026

Relevant Courses:

- Software engineering
- Concepts of Prog
- DS & Algo
- Databases
- Operating systems
- Data science
- Machine learning
- Deep Learning

Award: Excellence Scholarship (1000$) - Department of Computer Science and Operations Research

EXPERIENCE

Research Intern — Montreal, QC
Ubisoft La Forge — June 2025 – Aug 2025
Supervisors: Prof. Houari Sahraoui and Dr. Doriane Olewicki

- Worked on RevMate, an LLM-based code review assistant developed by Ubisoft.
- Implemented a systematic evaluation pipeline to quantify how effectively the tool reproduces human code review comments, using strict matching rules and LLM-as-a-judge similarity assessments.
- Developed an automated categorization pipeline that uses LLM-as-a-judge to classify generated comments into code review comments types, to identify coverage gaps across categories to guide RevMate's improvement.
- Conducted comparative evaluations to assess the impact of adding advanced techniques to RevMate (e.g., RAG and RAG with additional synthetic data) on the relevance of the generated code review comments.

Undergraduate Research Assistant — Montreal, QC
GEODES group - Université de Montréal. — Sept 2024 – Jan 2025, Jan 2026 – present
Supervisor: Prof. Houari Sahraoui

- Developed methods to simplify cluttered reverse-engineered UML class diagrams into higher-level conceptual representations.
- Used SBERT embeddings and LLM-based evaluation to distinguish domain-level concepts from implementation-level details.
- Implemented automation scripts for XML parsing, semantic similarity analysis, and pipeline orchestration.
- Ran Llama-70B-INT4 locally using ExLlamaV2 for large-scale evaluation experiments.
- Currently contributing to the project by constructing the ground-truth dataset and participating in technical discussions with the research team to evaluate implementation choices and potential refinements toward paper submission.

Teaching Assistant - Software Engineering (IFT2255) — Montreal, QC
Université de Montréal — Sep 2025 – Dec 2025

- Led weekly lab sessions supporting 50+ students in software engineering concepts, and held office hours to answer students' questions and guide them in semester projects.
- Graded assignments and projects, and provided clear feedback.

PROJECTS

Backtesting Tool for Trading Strategies

- Implemented a Python-based backtesting tool using Backtrader, NumPy, and Matplotlib to evaluate trading strategies.

Hockey goal prediction

- Built an expected goals (xG) prediction model for NHL shots using XGBoost, including data cleaning, feature engineering, and model evaluation.

Camouflaged Animal Detection System

- Designed a hybrid detection approach leveraging motion cues and CLIP embeddings to detect camouflaged animals in complex scenes, using the MoCA dataset.

EXTRACURRICULAR ACTIVITIES

Member of Escouade Millenium — Montreal, QC
Millenium Quebecor — Jan 2025 – present

- Representing Millennium Quebecor at events and engaging with students and professionals to promote entrepreneurship, innovation, collaboration, and the incubator's impactful programs.

SKILLS

- **Programming languages:** Java, Python, C/C++, Haskell, JavaScript, TypeScript, HTML/CSS, XML, SQL.
- **Libraries and frameworks:** NumPy, Pandas, Matplotlib, PyTorch, Backtrader, ExLlamaV2, Express.js, React, Node.js.
- **Tools:** GitHub, Git, IntelliJ, VSCode, PyCharm, Visual Paradigm, LaTeX.
- **Languages:** English · full professional proficiency | French · native | Arabic · native